

KGScope: Interactive Visual Exploration of Knowledge Graphs With Embedding-Based Guidance

Chao-Wen Hsuan Yuan , Tzu-Wei Yu , Jia-Yu Pan , and Wen-Chieh Lin 

Abstract—Knowledge graphs have been commonly used to represent relationships between entities and are utilized in the industry to enhance service qualities. As knowledge graphs integrate data from a variety of sources, they can also be useful references for data analysts. However, there is a lack of effective tools to make the most of the rich information in knowledge graphs. Existing knowledge graph exploration systems are ineffective because they did not consider various user needs and characteristics of knowledge graphs. Exploratory approaches specifically designed to uncover and summarize insights in knowledge graphs have not been well studied yet. In this article, we propose KGScope that supports interactive visual explorations and provides embedding-based guidance to derive insights from knowledge graphs. We demonstrate KGScope with usage scenarios and assess its efficacy in supporting the exploration of knowledge graphs with a user study. The results show that KGScope supports knowledge graph exploration effectively by providing useful information and helping explore the entire network.

Index Terms—Interactive visual exploration, knowledge graph, knowledge graph embedding.

I. INTRODUCTION

A KNOWLEDGE graph (KG) is a data model that represents the abundant relational information among real-world entities. There are several well-known knowledge graphs, e.g., Google Knowledge Graph [1], DBpedia [2], and YAGO [3]. Recently, a growing number of knowledge graphs have been constructed from specific domains [4], [5], [6]. They have become important resources to support applications in the retail, entertainment, education, and healthcare industries [4], [5], [6].

In addition to leveraging knowledge graphs for more accurate recommendations or better service quality of applications, information in knowledge graphs can also be useful for data analysts

to derive and reason about data insights [7], [8], [9], [10]. Nowadays, data to be analyzed can come in a variety of forms and sources, and it is difficult for users to explore such diverse data conveniently. Knowledge graphs can be used to integrate and represent data from different sources and express information effectively. However, knowledge graphs encode information into triplets of (*head*, *relation*, *tail*), which are originally designed to be processed by computing systems and are not intuitive for human users to consume.

Existing knowledge graph exploration systems typically offer a query box or search interfaces to start the exploration, assuming that users have a certain level of understanding of the data or have specific objectives in mind. Also, these systems often lack graph-centric analyses and only provide conventional analytical methods such as statistical filters [7], which may not be sufficient to reveal insights in knowledge graphs consisting mainly of semantic information. Although some work has proposed visualization systems [11], [12], [13] to visualize the graph structure, they have not leveraged entity embeddings to reveal semantic similarities between entities or provide proper guidance for exploring the graph. Therefore, there is a need for more advanced systems that can guide users in exploring knowledge graphs more effectively and provide more in-depth analyses tailored to the unique characteristics of knowledge graphs.

To make the most of the knowledge graph, we adopt knowledge graph embeddings to provide guidance to explore knowledge graphs (Fig. 1). Knowledge graph embeddings are vector representations of the entities and relations in a knowledge graph to simplify the manipulation while preserving the semantic meaning of the knowledge graph. We leverage these embeddings to develop three types of guidance for users: (a) peculiarity, (b) relation recommendation, and (c) similar neighbors. In addition to guidance, we use several visual designs and interactions, including brushing and linking, to support the entire workflow for users to draw insights from knowledge graphs (Section VI). Overall, our system, KGScope, provides a comprehensive set of tools and techniques to explore knowledge graphs that leverage the power of knowledge graph embeddings. These techniques enable users to identify and analyze relationships in the graph that may not be immediately apparent from its structure, making it a valuable tool for analysts.

We demonstrate the usage and generalizability of KGScope with different types of knowledge graphs in usage scenarios (Section VII) and validate its efficacy by comparing it with a baseline without embedding-based guidance in the user study

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Chao-Wen Hsuan Yuan, Tzu-Wei Yu, and Wen-Chieh Lin are with the College of Computer Science, National Yang Ming Chiao Tung University, Hsinchu 30010, Taiwan (e-mail: starlightwinnie@gmail.com; ap9940506@gmail.com; wcln@cs.nctu.edu.tw).

Jia-Yu Pan is with Google Inc., Mountain View, CA 94043 USA (e-mail: jiayu.pan@gmail.com).

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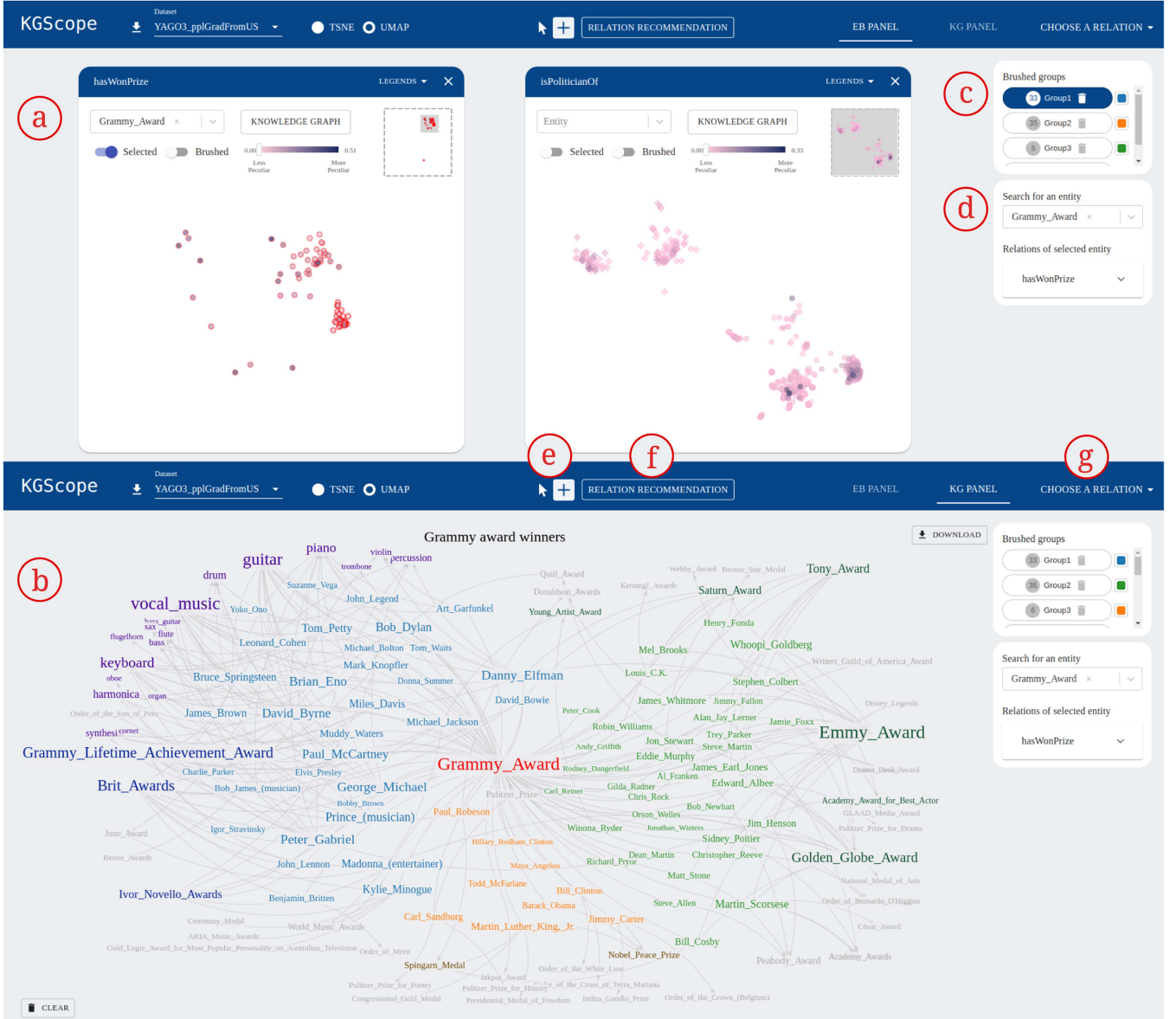


Fig. 1. Interface of KGScope. (a) Embedding panel. (b) Multi-relation knowledge graph panel. (c) Brushed groups panel. (d) Related relation panel. (e) Brush tool. (f) Relation Recommendation button. (g) Schema panel button. In this particular example, the user found that Grammy award winners could be categorized into three groups. Most of the people in Group 1 (blue) have a musical role and have won Grammy Lifetime Achievement Award, Ivor Novello Awards, and Brit Awards, which are all related to music. Many people in Group 2 (green) have won Emmy Awards, Tony Awards, and Golden Globe Awards related to TV, movies, and drama. Group 3 (orange) is composed of politicians.

(Section VIII). The results of the behavioral analysis and subjective feedback of the participants show that KGScope effectively supports the explorations of knowledge graphs in two aspects: (1) aiding the participants in conducting comprehensive and breadth-oriented explorations; (2) providing useful information to help the participants understand the data and guide them to find interesting insights.

Concretely, this paper makes the following contributions: (1) We propose KGScope which aims to enable data analysts to explore knowledge graphs and extract insights. (2) We introduce and demonstrate the effectiveness of knowledge graph embedding in visual analytics of knowledge graphs. (3) Exploiting knowledge graph embedding, we design metrics for measuring

the peculiarities of entities and recommending relations to guide users in exploring knowledge graphs.

II. RELATED WORK

A. Visual Exploration of Knowledge Graph

Visual exploration has been applied to various types of graph data, such as networks [14], [15], [16], large-scale graphs [17], multivariate and heterogeneous graphs [18], [19], [20], [21], [22], [23], and linked data [24], [25]. However, these works are not specifically tailored to knowledge graphs, nor do they leverage the properties of knowledge graphs to provide guidance in exploration.

To visually explore knowledge graphs, several systems [11], [12], [13] adopted graph-based visualizations to explore knowledge graphs due to their underlying structure [24]. *KGviz* [11] is a knowledge graph visualization framework that takes non-expert users into account by providing an intuitive user interface with manageable options. Users can search for and filter entities in knowledge graphs and visualize the result as a force-directed graph. *LodLive* [12] allows users to incrementally browse a knowledge graph, starting from an entity entered by the user, and then the user can see some information about the entity or can expand other entities connected to it, which are encoded by different shapes and colors to suggest the type of relation. *RelFinder* [13] automatically extracts the relationships between two entities of user interest in the knowledge graph and visualizes them as a force-directed graph with support for filtering, highlighting, and selecting. These methods concentrate on visualizing the structure of knowledge graphs for browsing without supporting in-depth analysis, such as accessing the relative similarities between entities, for users to derive insights efficiently.

Several works target visual analysis of knowledge graphs in specific domains. *ALOHA* [26] facilitates consumers to search for knowledge related to dietary supplements, designed exclusively for dietary supplement knowledge. It employs graph-based visualizations to help consumers understand dietary supplement information; specifically, *ALOHA* supports tasks such as identifying what ingredients a specific dietary supplement product contains and determining what drugs a dietary supplement product could interact with. *EduVis* [10] utilizes specific visualizations such as forced-directed graphs, timelines, and ring layout that are particularly suited to representing educational data and relationships. For example, the timelines in *EduVis* are designed to present the temporal data of the educational knowledge graph. As these works strive to deal with KGs of specific domains for users with explicit purpose, the designed features are mainly applicable to specific knowledge graphs, and did not consider the generalizability of the system. In contrast, *KGScope* is designed without dependence on the particular properties of specific knowledge graphs. The embedding-based guidances that *KGScope* provide for visual exploration are generally applicable on most knowledge graphs and also a major difference with *ALOHA* and *EduVis*.

B. Data Exploration Through Embedding Visualization

Several systems have been designed to interactively analyze particular embeddings [27], [28], [29], [30], [31]. Two recent papers visualize graph neural networks (GNNs) to help developers understand GNNs. *CorGIE* [27] and *GNNLens* [28] consist of multi-views, such as projecting the position of nodes in the latent space to 2D by t-SNE [32] or UMAP [33], and node-link diagrams to visualize the topological structure of the global and local area of the graph. *EmbeddingVis* [29] proposes an exploratory visual analytics system that supports the comparative visual interpretation of embedding vectors at multiple levels. These works explored the combination of multiple views

to better explore graphs from different perspectives, including graph space and embedding space.

The aforementioned methods are mainly designed to assist machine learning researchers in understanding the embedding models. They aim to find insights about embedding by analyzing projection distortions and evaluating what the model learns. In contrast, *KGScope* exploits graph embeddings to provide guidance to users in exploring knowledge graphs and extracting insights about the data itself.

C. Guidance Approaches in Visual Analytics

To explore the complex structure of knowledge graphs, users require guidance mechanisms that help them navigate the data and discover meaningful insights. The need for efficient and intuitive guidance mechanisms has led to various approaches.

Ceneda et al. [34] provides a comprehensive review of guidance approaches in visual data analysis. Several approaches discussed in the review paper [34] offer insights into guidance mechanisms that can be adapted for the exploration of knowledge graphs. For example, Streit et al. [35] designed a model to guide exploratory analysis, presenting alternative steps to take in order to complete a given task. Gladish et al. [36] developed an approach to show exciting data regions to explore during the analysis of hierarchical data, which could be applied to guide users through the exploration of knowledge graphs. These examples demonstrate the diverse range of guidance approaches, providing valuable insights for developing guidance in knowledge graph exploration.

Based on the definitions of Ceneda et al. [34], [37], *KGScope*'s approaches, such as embedding-based visualizations and the "peculiarity" measure, provide guidance that combines elements of *Orienting* and *Directing*. The system offers a high-level overview of knowledge graphs and suggests potential targets of interest to users, fitting the orienting guidance. It also provides a set of similar results through features like relation recommendation with preview, aligning with directing guidance.

III. DESIGN REQUIREMENTS

Based on previous studies and surveys on visualization and knowledge graph exploration [7], [38], [39], [40], [41], [42], [43], we identify a set of design requirements for *KGScope*.

R1. Provide guidance to extract findings: Knowledge graphs incorporate data from various domains, making them large and complex. As a result, the content can be difficult for users to grasp and familiarize themselves with [43]. As suggested in [7], a knowledge graph exploration system should be able to guide the user in their exploration journey to the interesting parts of the knowledge graph. Li et al. [44] also pointed out the need for the ability to immediately begin analysis at the user's region of interest in the knowledge graph. Moreover, the latent information behind the knowledge graph is useful for exploration, but is not explicitly visible to users. Providing guidance that reveals and leverages the latent information enables users to uncover implicit relationships and navigate through the complex contents with ease.

R2. Ensure awareness of the context: In knowledge graph visualization, users often face the challenge of Viewability [11]. Displaying a large graph in limited space causes nodes and edges to overlap with each other, whereas focusing on a part of the graph prevents users from seeing the contextual information of entities and relations. Recent research has pointed out the need for maintaining context in knowledge graph visualization [44]. Also, during the exploration process, new meanings can be brought to the observation with context information at hand, and this is when insights occur [41]. To facilitate the exploration process and reduce cognitive load, the systems should help users keep track of the overall context [38].

R3. Facilitate analysis from different perspectives: Knowledge graphs are characterized by their multi-relational structure, which can overwhelm users when numerous entities and their diverse relations are displayed simultaneously [38]. In particular, entities can form distinct relative associations when being considered in the context of different relations; therefore, combining various relations of the entities in a view may obscure information details. To help users gain a deeper understanding of complex information and uncover patterns that may not be apparent from a single perspective [31], systems should allow users to view entities from the perspective of separate relations and support comparison between multiple relations [39]. Enhancing this capability to switch views while maintaining context would further empower users to analyze the graph from multiple perspectives without losing sight of the overall context [44].

R4. Promote analysis in groups: In addition to inspecting individual entities, analyzing entities in groups is a common technique to discover insights, such as surprising clusters or important groups [40]. For users who do not have an idea in advance, through observing the data, they may gain interest in the cluster of entities grouped in the visualization. Moreover, users may also want to compare multiple groups of entities. Effective exploration tools should allow users to select groups of entities on demand, reveal possibly related information, and reduce irrelevant data [44].

IV. BACKGROUND

In this section, we introduce the background of knowledge graph embeddings (Section IV-A) and link prediction (Section IV-B) to facilitate the understanding of our proposed approach.

A. Knowledge Graph Embedding

A knowledge graph is a multi-relational graph consisting of entities and relations. Entities are represented as nodes, and relations are represented as edges. Each edge represents a fact of a relation between two entities and can be denoted as a triplet (h, r, t) , where h is a head entity, r is a relation and t is a tail entity. For example, the triplet $(Paris, capitalOf, France)$ indicates that Paris is the capital of France.

To facilitate the exploration of knowledge graphs, we propose to exploit knowledge graph embeddings [45], [46]. The main idea of knowledge graph embeddings is to represent entities and relations in a continuous vector space, so as to simplify the manipulation while preserving the semantic information of the

knowledge graph. The ability of knowledge graph embeddings to quantify the relationships between entities is something that knowledge graphs themselves do not possess.

A typical knowledge graph embedding technique generally consists of three steps: (1) representing entities and relations, (2) defining a scoring function, and (3) learning entity and relation representations [45]. Here, we illustrate these steps using TransR [47], one of the models we employ in our work, as an example. In TransR, entities are represented as vectors in an entity space $\mathbb{R}^d$, and each relation is associated with a specific space $\mathbb{R}^k$ and is modeled as a translation vector in that space. The vectors of entities and relations can be initialized with random values or from the results of other simple models like TransE [48]. Given a triplet (h, r, t) , TransR first projects the representations of the entities h and t into the space specific to relation r , i.e.,

$$h_{\perp} = M_r h, t_{\perp} = M_r t$$

Here $M_r \in \mathbb{R}^{k \times d}$ is a projection matrix from the entity space to the relation space of r . Then, the scoring function that measures the possibility of a triplet's existence is defined as

$$f_r(h, t) = -\|h_{\perp} + r - t_{\perp}\|_2^2$$

where a higher score indicates that the triplet (h, r, t) is more likely to be a fact. Finally, TransR applies the following margin-based loss function as the objective for training

$$L = \sum_{(h, r, t) \in \mathcal{S}} \sum_{(h', r, t') \in \mathcal{S}'} \max(0, f_r(h, t) + \gamma - f_r(h', t'))$$

where γ is the margin, $\mathcal{S}$ is the set of correct triplets and $\mathcal{S}'$ is the set of incorrect triplets. For different knowledge graph embedding models, their score functions are defined differently.

B. Link Prediction

Link prediction [47], [49] is the task of predicting an entity given a relation and another entity, e.g., predicting a head entity h by giving a relation r and a tail entity t , denoted $(?, r, t)$, and vice versa.¹ For example, in the case of tail prediction, given a truth triplet (h, r, t) , the quality of an embedding model in modeling the triplet can be evaluated using the *prediction rank*. To compute the rank of the ground-truth entity t , every other tail entities t' is substituted in the triplet (forming a candidate triplet (h, r, t')) and the corresponding score is computed. Then the rank of each candidate is obtained by sorting the candidates based on their scores in descending order. Ideally, if the rank of (h, r, t) is ranked before any other (h, r, t') , it means that the computed graph embedding model is good.

After computing the rank of each triplet, the mean rank is a common metric used to evaluate the quality of a graph embedding model for link prediction, which is simply the average of the predicted ranks. It should be noted that the interpretation of the mean rank depends on the number of candidates. For example,

¹The term Link Prediction has also been defined differently in [50], [51]. However, the definition we adopted in this paper is a more common definition in the field of knowledge graph embedding models.

a mean rank of 2 might indicate great performance with 1,000 candidates, but extremely poor performance with 2 candidates.

In the next section, we will introduce our proposed approach for identifying peculiar entities using the rank of link prediction results. It is important to note that our approach does not involve predicting new triplets outside of the existing knowledge graph. Instead, we utilize link prediction to measure the rank of triplets within the graph, with the goal of identifying entities that exhibit unexpected behaviors.

V. PECULIARITY GUIDANCE

Unexpectedness was identified as one of the seven characteristics of data insights [41]. Unexpected insights may challenge preconceived notions or assumptions about a dataset or phenomenon and can provide valuable opportunities for learning and discovery in data analysis. In this work, we propose *peculiarity* as a measure of unexpectedness for entities or relations and use it as exploration guidance to reveal unexpected information. The idea behind the peculiarity is that assuming that knowledge graph embeddings have captured the common patterns or behaviors of the majority, and that an entity e is *peculiar* if the results of link prediction of e are inaccurate, that is, whether there are other entities ranked before e . In other words, we measure the peculiarity based on the number of false answers that are ranked before the true ones. Intuitively, if the predicted rank of an entity e is lower than expected, then e could be more peculiar.

Let $Rel(e)$ be the set of all relations involved with an entity e in a KG . If e is a head entity in a relation $r \in Rel(e)$, then $\mathcal{T}(e, r) = \{t_i | i \in [1, T_{e,r}]\}$ is the set of the tail entities that e links to in r , where $T_{e,r}$ is the number of entities in $\mathcal{T}(e, r)$. For every tail entity t' in r , the score of (e, r, t') is calculated based on the score function of a given knowledge graph embedding eb . The rank of each candidate t' , denoted as $R_{eb}^{(t)}(t'|e, r)$, where (t) indicates tail prediction, is obtained by sorting the candidates based on their score in descending order.

As an entity may link to many entities, ranking any of these entities at the front of the list should be considered correct. As an example, if an advisor p advises two students, then the embedding model inevitably has to rank one student after the other when predicting the correct entity of $(?, hasAdvisor, p)$ even when both are equally correct. In this case, the rank of the lower one is unfairly computed, which could affect the estimation of peculiarity. Hence, to fairly compute the rank of an entity, the *filter setting* [49] is usually applied to disregard other correct candidates when calculating the rank. We call the rank that applies the filter setting as *adjusted rank* and denote the adjusted rank of (e, r, t_i) predicted by eb as:

$$\hat{R}_{eb}^{(t)}(t_i|e, r) = R_{eb}^{(t)}(t_i|e, r) - \Delta R_{eb}^{(t)}(t_i|e, r), \quad (1)$$

where $\Delta R_{eb}^{(t)}(t_i|e, r)$ is the number of tail entities of e that are ranked before t_i . Formally, $\Delta R_{eb}^{(t)}(t_i|e, r) = |\{t_j | j \in [1, T_{e,r}] \text{ s.t. } R_{eb}^{(t)}(t_j|e, r) < R_{eb}^{(t)}(t_i|e, r)\}|$ and $|\{\cdot\}|$ is the cardinality of a set. Similarly, when e is a tail entity in relation r that links to $H_{e,r}$ head entities, $\mathcal{H}(e, r) = \{h_i | i \in [1, H_{e,r}]\}$, the adjusted rank of (h_i, r, e) when performing head prediction

TABLE I
CHARACTERISTICS OF THE EMBEDDINGS USED IN THIS WORK

Embeddings	Space	Operations
TransR	Euclidean	Translation
RotatE	Complex	Rotation
AttH	Hyperbolic	Rotation & Reflection

TABLE II
CORRELATIONS BETWEEN THE PECULIARITIES OF ENTITIES GIVEN BY DIFFERENT EMBEDDINGS

Dataset	Embeddings	r
YAGO-US-Graduates	TransR/RotatE	0.1253
	RotatE/AttH	0.4224
	AttH/TransR	0.2901
TMDB-After-2000	TransR/RotatE	0.1825
	RotatE/AttH	0.4351
	AttH/TransR	0.3735

R indicates the pearson correlation coefficient between the entities' peculiarities of two embeddings, e.g., r between trans and rotate is 0.1253. The results suggest that any two embeddings are correlated and complement one another.

by eb is:

$$\hat{R}_{eb}^{(h)}(h_i|e, r) = R_{eb}^{(h)}(h_i|e, r) - \Delta R_{eb}^{(h)}(h_i|e, r), \quad (2)$$

where $\Delta R_{eb}^{(h)}(h_i|e, r)$ is the number of head entities of e that are ranked before h_i . The peculiarity of e in r is defined as follows:

$$P_{eb}(e, r) = \frac{1}{T_{e,r} + H_{e,r}} \times \left(\sum_{i=1}^{T_{e,r}} \frac{1}{T_r} \hat{R}_{eb}^{(t)}(t_i|e, r) + \sum_{i=1}^{H_{e,r}} \frac{1}{H_r} \hat{R}_{eb}^{(h)}(h_i|e, r) \right), \quad (3)$$

where T_r and H_r are the total number of tail and head entities in r , respectively. The peculiarity measure of e considers the ranks of its heads and tails. Because the numbers of heads and tails in r may not be the same and the interpretation of the ranks depends on the number of candidates, the adjusted ranks are normalized before averaged.

In our design, we recognize that different embedding models put emphasis on encoding different kinds of relationships between entities; therefore, we utilize a set of embedding models $EB = \{TransR[47], RotatE[52], AttH[53]\}$, to capture patterns from various perspectives in a knowledge graph and discover truly peculiar entities. The characteristics of each embedding model are shown in Table I, and the correlations between the peculiarities of entities given by different embeddings are shown in Table II.

Our proposed design is expansible, and other embedding models can also be included in the set. However, when it comes to modeling connection patterns in knowledge graphs such as symmetric, inversion, and composition [52], using knowledge graph embeddings intuitively offers better quality compared to general graph embeddings, as these specific connection patterns may not exist in general graphs. The research

TABLE III
EXAMPLES OF NODES WITH HIGH OR LOW PECULIARITIES FROM
YAGO-US-GRADUATES KG

High Peculiarity Examples		Low Peculiarity Examples	
Entity Name	Peculiarity	Entity Name	Peculiarity
Imee Marcos	2.666	Michael Walzer	0.012
Chelsea Clinton	2.522	Paul Bowles	0.011
Barack Obama	2.485	David Eddings	0.010
Amanda Palmer	2.424	Hermann Hesse	0.009
Madonna (entertainer)	2.421	Allen Tate	0.008

community has developed specialized embedding methods for knowledge graphs to make better use of information in knowledge graphs.

To take into account every embedding model, we sum up $P_{eb}(e, r)$ for all eb in the set EB to obtain the peculiarity of e in r , i.e. $P(e, r) = \sum_{eb \in EB} P_{eb}(e, r)$.

From an exploratory perspective, all the information related to an entity should be considered when evaluating the peculiarity of the entity. Thus, we sum up $P(e, r)$ for all relations linked to e to obtain its peculiarity, i.e., $P(e) = \sum_{r \in Rel(e)} P(e, r)$. One may think that $P(e)$ could cause the bias that an entity with more relations tends to have a higher peculiarity. However, an entity with more relations, indeed, also contains more information. Therefore, such an entity still deserves a higher peculiarity.

To illustrate the concept of peculiarity, Table III gives examples of nodes with high or low peculiarities in YAGO-US-Graduates KG, which is a KG used in Section VII. After inspection, we found that the high peculiarity of Barack Obama comes from the fact: (*Barack Obama, Has Won Prize, Grammy Award*), which is unusual for politicians. Another example of a highly peculiar entity in the relation *Has Won Prize* is Ben Stein, a person with a diverse career as an actor, writer, lawyer, and commentator, who also won an Emmy Award which is typically awarded to people with a career in the entertainment industry. For low-peculiarity entities, an example is given by David Eddings. The triplets related to David Eddings are (*David Eddings, Was Born In, Spokane Washington*), (*David Eddings, Died In, Carson City Nevada*), and (*David Eddings, Has Gender, Male*), which are basic information of a person and nothing special. Other similar low-peculiarity entities are also listed in the table.

VI. PROPOSED SYSTEM: KGSCOPE

In this section, we introduce the proposed solutions in KGScope that fulfill the design requirements identified in Section III. To provide the context during the exploration, the *schema panel* (Figs. 1(g) and 2) exhibits the schema with visualizations that encode overview statistics of the knowledge graph, and the *EBView* (Figs. 1(d) and 3) visually encodes the relative associations between entities in a relation (Section VI-A). To offer useful hints to guide the user exploration, we quantify the *peculiarity* of entities and color-encode the entities accordingly in the EBViews to disclose informative directions for exploration (Section VI-A). When a user is interested in a set of data, KGScope reduces irrelevant information with the filtering and brushing tools (Fig. 1(c), Section VI-B); and

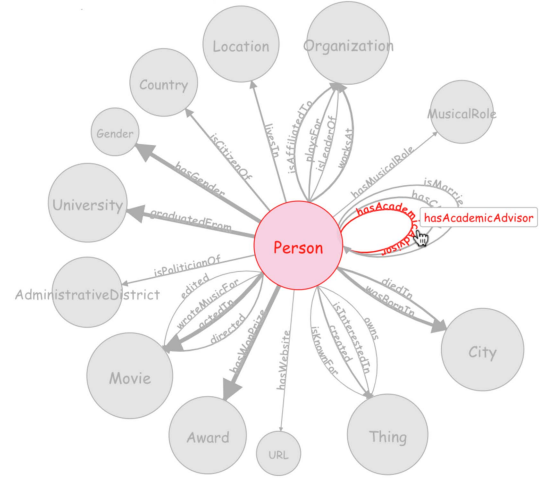


Fig. 2. Hovering on the relation, “hasAcademicAdvisor”, in the schema panel of YAGO-US-Graduates.

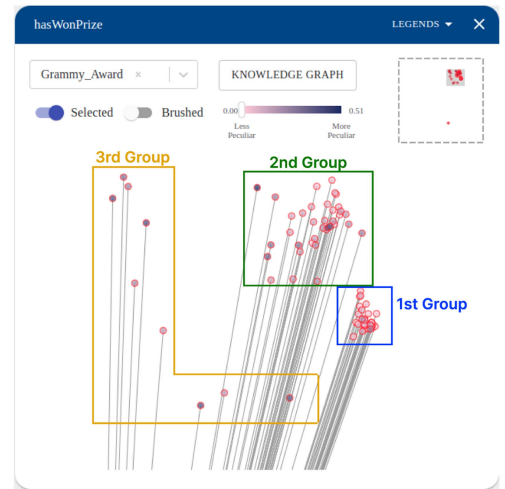


Fig. 3. Scenario 1 - The distributions of Grammy Award awardees in the EBView.

also recommends relations based on the embedding (Fig. 1(f), Section VI-C). To facilitate users to derive insights from an exploration, the KGView (Fig. 4) is provided to assist users in interpreting the relationships between entities intuitively and the MRKGPanel (Fig. 1(b)) can integrate the observations obtained during an exploration for users to summarize their findings (Section VI-D). The implementation details can be found in Appendix A, available online.

It's worth noting that although visual encodings in KGScope are not specific to knowledge graphs, all the other features, such as peculiarity, relation recommendation, and similar neighbors are designed specifically for KGs.

A. Context of Knowledge Graphs

KGScope visualizes two kinds of information to provide the context of a knowledge graph. First, visualizing the entity-relation schema (Fig. 2) gives users a high-level understanding of the structure of the knowledge graph, with visual hints about the

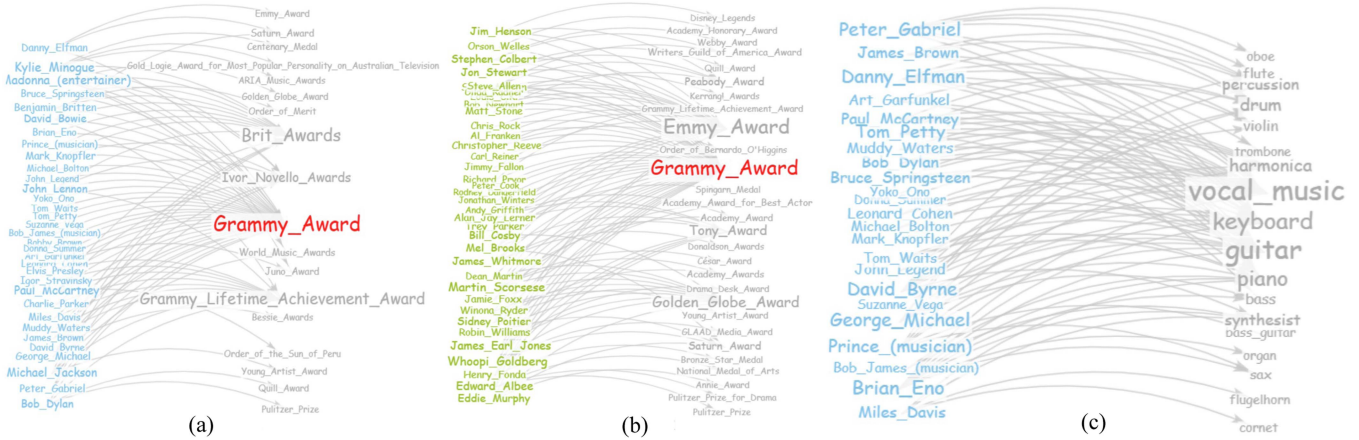


Fig. 4. Scenario 1: From the knowledge graph views, we can find other entities or relations connected with the entities in different groups, e.g., other awards that Grammy Awardees won or their other relations. (a) In Group 1, some people also won Brits Awards and Ivor Novello Awards. This might imply that Group 1 is made up of musicians. (b) In Group 2, some people also won an Emmy Award and a Golden Globe Award. (c) In relation “hasMusicalRole”, people in Group 1 have roles like vocal music, keyboard, guitar, etc. This could further confirm that Group 1 is musicians.

statistics related to the relations and entities. We choose to use a force-directed graph to preserve ontological insights, similar to its usage in VOWL 2 [54] for visualizing ontologies, instead of other graph representations, such as the adjacency matrix. In the schema panel, a circular node represents an entity type and its size encodes the number of entities belonging to this type; a directed edge pointing from a head node to a tail node represents a relation, and the edge width encodes the number of triplets in this relation. The schema panel helps users get an overview of the underlying data (R2), and guides the user to areas of potential interest to start the exploration (R1).

The second proposed visual design to display the context during a data exploration is the proposed *embedding view* (EBView). To facilitate analysis from different aspects of the context, we adopt the knowledge graph embedding model, TransR [47], due to its ability to learn semantic information and model entities in euclidean space. One particular property of the TransR model is that, for each relation, an entity is mapped to a different vector space, giving the model more flexibility in capturing the different data perspectives of the various relations. Then, for each relation, we employ the dimension reduction technique, UMAP [33], to project the high-dimensional embedding vectors into a two-dimensional space and visualize them in a scatter plot that we call EBView (Fig. 3). In EBView, the entities are color encoded according to the peculiarity (Section V). Head and tail entities are represented as circles and crosses, respectively. If an entity is both a head and a tail, it is represented as a triangle.

In KGScope, there are two ways to open the EBView of a relation of interest. The first way is to choose a relation directly from the schema panel, and the second way is to select from the *related relation panel* (Fig. 1(d)). The related relation panel shows the relations in which an entity e is involved when e is selected, and the relations are sorted by peculiarity $P(e, r)$.

The contextual information provided by EBView can also act as a type of guidance, similar neighbors, for users to obtain semantically similar or dissimilar entities by observing the

relative distances between them (R1), and the use of EBView allows analysis from different perspectives (R3).

B. Group Analysis

KGScope supports exploration tools for three types of groups: (1) *keyword group* formed by entities with similar keywords, which can be found by the search box in KGScope; (2) *sibling group*, composed of entities linked to a common entity in a relation, can be identified in KGView after choosing an entity; (3) *visual group*, formed by the entities that are grouped by the EBView (Fig. 3), which can be manually formed using the brushing tool. In addition, various groups can be formed in EBView with the focusing and filtering tools provided in KGScope. To exclude irrelevant information in groups, users can zoom in, turn on the toggle to show only the selected/brushed entities and the entities linked to them, or filter out entities with peculiarities lower than a specified threshold.

To analyze a group of entities from different perspectives, multiple EBViews are displayed simultaneously so that users can handily compare between relations (R3). The EBViews are accompanied by a brushing-and-linking tool, with which, when selecting a group of entities in one view, the selected entities in the other views will also be highlighted (Fig. 5(b)). The reason we place the *brush mode* button in the top bar is to be able to distinguish brush from select. Without this explicit toggle UI, we found that it would make users forget that they could brush. An alternative design is to change the mouse cursor into the crosshair shape when the user's mouse enters the EBView; however, this design makes users forget that they can select an entity reversely. KGScope establishes a group-friendly environment to assist users in probing into entity groups (R4).

C. Relation Recommendation by Contextual Similarity

As KGScope allows and promotes the analysis of entity groups, recommendation of (interesting) relations for a group of entities is also supported. Since the distance between two

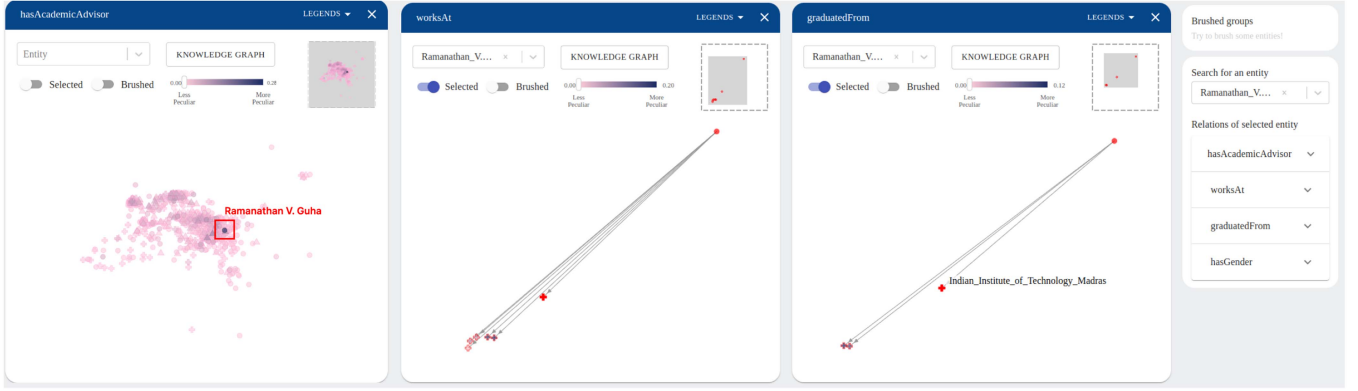


Fig. 5. Scenario 2. (a) The entity “Ramanathan V. Guha” is first identified for its relatively high peculiarity. (b) After selecting the entity, we can inspect the EBViews of other relations that connected to it from the related relation panel, in which “worksAt” and “graduatedFrom” are ranked among the top three peculiar relations (“hasAcademicAdvisor” ranked first).

entities in a knowledge graph embedding reflects their similarity, the embedding distance is utilized to measure similarity. Instead of directly measuring similarity within a group of entities G , we propose *contextual similarity*, which measures the similarity of other entities connected to G . This is based on the concept that *the characteristics of each entity in a knowledge graph are defined by the connections related to it* and can assist users in *finding the commonalities efficiently*. The similarity score of the entities in G in relation r is defined by

$$S_G(r) = - \left(\sum_{\forall x, y \in G', x \neq y} d_r(x, y) \right) / \left(\frac{|G'| \times (|G'| - 1)}{2} \right), \quad (4)$$

where $d_r(x, y)$ is the euclidean distance between entities x and y in the graph embedding of r ; G' denotes all the entities connected to G in relation r ; $|G'|$ is the number of entities in G' . The recommended relations are presented with thumbnails of EBView rendered and sorted by similarity scores in descending order, which facilitate comparison between multiple relations (R3).

The relation recommendation offered users potentially interesting relations for a group of entities (R4), which can also serve as guidance to guide users to the interesting part (R1).

D. Interpretation and Summarization of Linking Patterns

Despite the efficiency of EBPanel in providing the global view to assist users in navigating through entities, the relationships between entities may not be intuitively interpreted in the panel. To better derive insights, each EBView is accompanied by a corresponding knowledge graph view (KGView) with force-directed layout [55] to reduce occlusion (Fig. 10). The size of the text label encodes the sum of the indegree and outdegree of an entity. KGScope offers another option to encode entities of different groups with different colors.

Furthermore, to summarize the observations during exploration, users can save the KGView they consider intriguing, and the subgraphs will be integrated into the multi-relation knowledge graph panel (MRKGPanel) (Figs. 1(b) and 4). Having a

panel that displays all the explored subgraphs has advantages, including allowing users to observe the connection between these subgraphs, such as the commonality among them (R3), and helping users to keep track of the overall context (R2).

VII. USAGE SCENARIOS

In this section, we describe two usage scenarios to demonstrate how KGScope meets user needs in exploring and discovering insights from knowledge graphs. More usage scenarios can be found in Appendix B, available online.

Data preparation: Yet Another Great Ontology (YAGO3) [3] is a large semantic knowledge base derived from Wikipedia, GeoNames and other data sources. It contains 37 relations and 123,182 entities, including people, organizations, etc. We extracted a subset of YAGO3 and created a knowledge graph (referred to as *YAGO3-US-Graduates*) that contains data from people who graduated from universities in the United States. YAGO3-US-Graduates includes all the triplets whose head or tail entities are either these people or other people who are related to them in any of the relations. In the end, we have 43,193 triplets with 15,632 entities and 25 relations, such as *graduatedFrom* and *hasWonPrize*.

A. Scenario 1: Group Analysis of Grammy Award Winners

We type in “Grammy Award” in the search box to start the exploration. By observing EBView of the relation “hasWonPrize”, we roughly identify three groups among the awardees (Fig. 3). We brush them separately and inspect them in KGView. Through interpreting the names of the entities, it is surprising that the entities in the third group are mainly politicians, and they also have higher peculiarity values. For the other two groups, we cannot immediately differentiate between them through their names; thus we inspect the other awards they have won.

The KGView of the first group (Fig. 4(a)) shows that other awards they have won are the Grammy Lifetime Achievement Award, the Ivor Novello Award and the Brit Award, which are related to music. For the second group, other awards they have won include the Emmy Award, Tony Award, and Golden

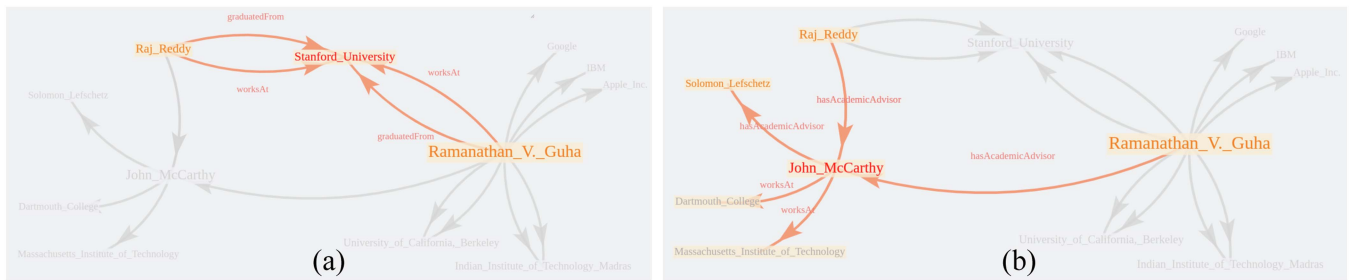


Fig. 6. MRKGPanel in usage scenario 2. (a) We hover over “Stanford University” and see that both “Raj Reddy” and “Ramanathan V. Guha” graduated from Stanford. (b) We hover over “John McCarthy” and see that he is the advisor of Reddy and Guha, but has no connection with Stanford.

Globe Award, which are related to the TV, theater, and film industry (Fig. 4(b)). It appears that the people in group 2 are actors. All views saved during exploration are summarized in the MRKGPanel (Fig. 1(b)).

Besides “hasWonPrize”, we also explore whether the 3 groups can be distinguished in other relations. Since we have no idea which relation to examine, rather than doing random explorations, we take advantage of KGScope’s relation recommendation feature and direct our exploration according to the recommended relations for the first group. We examine the relation “hasMusicalRole”, which is ranked second in the recommendation with 23 entities in it. The people in the first group have musical roles such as vocal music, keyboard, guitar, etc. (Fig. 4(c)). When we switch to the second group, we find that none of them has a musical role. This again confirms our previous findings that the Grammy Award winners can be divided into 3 groups: musicians, actors, and politicians. The colors in MRKGPanel allow us to recognize different groups and wrap up our findings efficiently.

Scenario 1 shows that KGScope facilitates group analysis and the relation recommendation enables users to find commonalities of a group, which may further distinguish between groups.

B. Scenario 2: Uncovering the Connection Between John McCarthy and Stanford University

We select the relation “hasAcademicAdvisor” from the schema panel. By viewing the EBView of “hasAcademicAdvisor” (Fig. 5(a)), the entity Ramanathan V. Guha, which has the highest peculiarity (in dark purple), immediately attracts our attention. After clicking the entity, we know that his advisor is John McCarthy. We further browse the related relation panel where relations are ranked based on the peculiarities. Since “hasAcademicAdvisor” is the first relation, we explore the second and third relation, which are where he works and graduated. We open the EBViews of both relations and explore them side by side (Fig. 5(b)). By hovering over the entities, we find that he also works at the three institutions he graduated from. Moreover, the two institutions in the United States, “Stanford University” and “University of California, Berkeley”, are close to each other; while “Indian Institute of Technology Madras” is relatively far from the others in the EBViews. This may imply

that the embedding could implicitly encode the geographical information of these entities. In addition to teaching at the universities, he also worked at IBM, Google, and Apple.

After inspecting Guha, we go back and select McCarthy. In the KGView of “hasAcademicAdvisor”, we perceive that McCarthy has another advisee, Raj Reddy, and an advisor, Solomon Lefschetz (Fig. 6(a)). We also examine the relation “worksAt” which is the second relation ranked based on the peculiarity in the related relation panel. We do not recognize any organizations at which he worked that seem to be where Guha graduated. Then, we begin to wonder where McCarthy advised Guha. By saving the views during exploration to the MRKGPanel, we discover that two links of the relations “graduatedFrom” and “worksAt”, pointing from Reddy to Stanford University, are also revealed (Fig. 6(a)). We assume that McCarthy advised them at Stanford University because it is the only entity commonly connected to them in the relation “graduatedFrom”. However, the knowledge graph only tells us that McCarthy worked at Dartmouth College and Massachusetts Institute of Technology (Fig. 6(b)); thus, we believe that there is a missing link of “worksAt” between McCarthy and Stanford University. To confirm our conjecture, we search on Wikipedia and find that he had a short-term appointment at Stanford before becoming an assistant professor at Dartmouth in 1955, and then in 1962, he became a full professor at Stanford, where he remained until his retirement in 2000.

This scenario shows that KGScope could disclose interesting information through peculiarity and support comprehensive exploration by revealing related relations. Furthermore, through interactive analysis, KGScope could even guide us in finding a missing link that may not be easily identified by automatic algorithms.

VIII. USER STUDY DESIGN

We conducted a user study with a comparative analysis to assess how the designs of KGScope can meet the design requirements. As existing systems [11], [12], [13] varied in goals, features, and representations, we implemented a baseline system to have a more controlled comparison. We focus on investigating how the behaviors and findings of the participants differ when exploration is performed with and without the guidance and features supported by KGScope.



Fig. 7. Interface of the baseline system.

A. Baseline

We designed a system that imitates existing knowledge graph exploration systems without embedding-based guidance as the baseline. The baseline is akin to the existing systems that mainly explore raw data without guidance or context. Similarly to an ablation study, this baseline allows us to isolate and evaluate the specific contributions of our system’s unique features by comparing them against a more basic model. In this way, we can directly observe how our proposed guidance features improve the user experience on KG explorations.

Fig. 7 shows the interface of the baseline. It is a simplified version of KGScope where the EBPannel, the relation recommendation panel, and the brush groups panel are removed. In addition, the drop-down menu of the entity search box and the relations in the related relation panel in the baseline are randomly ordered. Users can incrementally expand the graph by clicking the nodes in the graph or selecting entities from the search box and then selecting the relations to expand. The knowledge graph data are also visualized as a force-directed graph. Finally, users can wrap up their insights in the same view where they explore the knowledge graphs.

B. Datasets

Besides the YAGO3-US-Graduates dataset, three other datasets were used to assess the applicability of KGScope in different domains.

TMDB-After-2000 is sampled from TMDB 5000,² which is a dataset of 5000 movies with metadata, such as casts and production companies. TMDB-After-2000 contains the top 20 main casts in the cast list of each movie and movies released after 2000 with revenues of more than \$250,000,000. It contains 30131 triplets with 12232 entities and 10 relations, including *hasKeyword*, *isProducedIn*, etc.

KG20C [56] is extracted from the Microsoft Academic Graph dataset [57]. It contains computer science papers published in top conferences between 1990 and 2010. There are 55607 triplets with 16362 entities and 5 relations, including *isAffiliatedTo*, *cites*, *isPublishedIn*, etc.

*Marvel KG*³ is extracted from the Marvel Database,⁴ which introduces the main characters in the Marvel Universe. There are 1283 triplets with 664 entities and 25 relations, such as *leads*, *competesWith* and *hasTitle*. This dataset was used only in the demo and in practice.

C. Participants

We recruited 12 postgraduate students (5 women and 7 men, average age 23.5) who are experienced in data analysis. The average self-reported familiarity with data exploration/analysis on a 5-point scale (1 for “no experience at all” and 5 for “doing almost every day”) was 3.5 ($\sigma=1.3$). All participants were screened to ensure that they had neither explored the study datasets before nor used the baseline and KGScope.

D. Procedure and Tasks

We adopted a within-subjects design in which each participant conducted an exploratory session with Baseline and another with KGScope, on the same dataset. *Marvel KG* was used for practicing and the other 3 datasets were evenly distributed among the 12 participants according to their preferences. In the end, each dataset was explored by 4 participants. The order of the systems each participant used was counterbalanced to avoid bias.

In the beginning, we briefly introduced knowledge graphs to the participants. For each session, we provided a video tutorial of the system using *Marvel KG* as the demo dataset. The participants then practiced using the features of the system with *Marvel KG*. After practicing for 10 minutes, two general tasks were given in each session to the participants in the formal study.

In Task 1, each participant conducted an open-ended exploration on the given dataset for 15 minutes. In Task 2, each participant was first asked to choose an entity e of his/her own interest and it should be connected to more than one entity in a relation r . Then the participant explored the entities linked to e in r for 10 minutes. This task was used to investigate a participant’s behavior in focused explorations of a sibling group. Participants were asked to think aloud during their explorations and summarize the insights they found by typing the descriptions into the systems at the end of each task. Participants were given a 10-minute break between two sessions. After completing the two sessions, participants completed the post-test questionnaires and had interviews. Each participant spent approximately 2 hours completing the experiment. The audio, screen, and interaction logs of the participants in both systems were recorded.

E. Post-Test Questionnaire and Interview Questions

The *post-test questionnaire* includes 5-point Likert scale ratings (1=“strongly disagree” and 5=“strongly agree”) of 6 questions regarding the usability of both systems and the rationale for the ratings.

- Q1. The system is applicable for open-ended exploration (Task 1).

²<https://www.kaggle.com/tmdb/tmdb-movie-metadata>

³<http://openkg.cn/dataset/marvel>

⁴<https://marvel.fandom.com/wiki/Category:Characters>

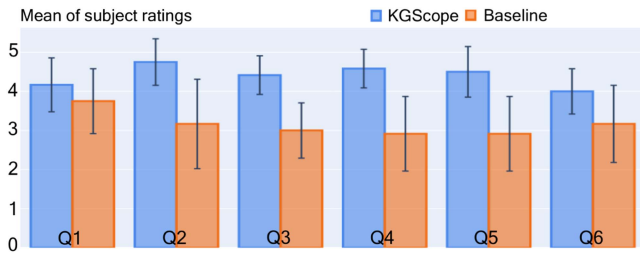


Fig. 8. Mean of participants' 5-point Likert scale ratings on the usability of KGScope and baseline (1="strongly disagree" and 5="strongly agree"). Q1. Open-ended exploration; Q2. Focused exploration; Q3. Usefulness of information; Q4. Efficiency of interpretation; Q5. Easiness of wrapping up; Q6. Insight satisfaction. KGScope received significantly higher ratings ($p < 0.05$) than the baseline in all aspects except Q1. The error bar denotes the standard deviation.

- Q2. The system is applicable for focused exploration (Task 2).
- Q3. The system provides useful information to assist the exploration of knowledge graphs.
- Q4. The system enables me to interpret the relationships between entities efficiently.
- Q5. The system enables me to easily wrap up the insights I found during the exploration.
- Q6. I am satisfied with the insights I found.

The *interview questions* are listed as follows:

- 1) Have you ever explored knowledge graphs or any similar graph data?
- 2) Please describe briefly your exploration process with the two systems. Did you apply any different strategies during the exploration with the two systems? Why?
- 3) To start the exploration, what would you like to know about the dataset? Why?
- 4) What do you think about displaying different relations in a view and separating views?
- 5) What things would you like to remember during the exploration? Why?
- 6) What kinds of insights do you think are more satisfying? Why?
- 7) Did you have any difficulties or something that made you confused during the exploration with the two systems? Or anything you wanted to do but the systems did not support?

IX. RESULTS

We report the user study results, including participants' questionnaire ratings, the insights they found, their behaviors during sessions, and their feedback in the interviews.

A. Usability From the Post-Test Questionnaire

As shown in Fig. 8, participants gave significantly higher ratings ($p < 0.05$) than the baseline on KGScope in all questions except Q1. They perceived that KGScope and the baseline offer similar support in the open-ended explorations (Q1), but KGScope obviously excelled at the focused explorations (Q2). Participants considered the information provided by KGScope useful to help them decide which parts of the data to explore

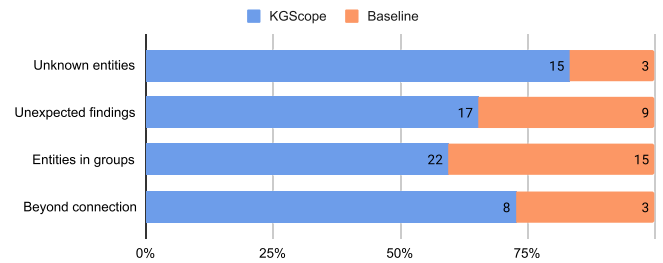


Fig. 9. Comparison of KGScope and Baseline: The number of insights found by participants related to the four categories.

(Q3), to interpret the relations between entities efficiently (Q4), and to wrap up the explorations with satisfying insights (Q5, Q6).

B. Behavioral Analysis Results

We analyzed the participants' 48 findings (12 participants $\times$ 2 systems $\times$ 2 tasks) with their interaction logs and screen recordings with audio input to capture "think aloud" feedback. Our analysis reveals that KGScope allows participants to: (a) be more willing to explore unknown entities; (b) obtain more findings that are unexpected to the users; (c) explore entities in group and uncover relevant and interesting insights; (d) derive insights that are beyond direct connections. Fig. 9 presents a comparison of the KGScope and Baseline, with the number of insights in each of four categories.

Participants were more willing to explore unknown entities in KGScope: We observed that although most participants tended to explore entities they had known, some insights included entities that were unknown to them. Of the 18 insights related to unknown entities, only 3 of them were found with the baseline, while the others were found with KGScope. This inlines with participants' feedback (Section IX-C) that they inspected the entities based on the relative associations and discrepancies between data. Revealing these hints in KGScope enabled them to be not restricted to the entities' names they had known.

Participants mentioned unexpected observations in their insights: As described in Section IX-C, speaking of satisfying findings, most of the participants came up with those that were unexpected to them. By analyzing their insights, we noticed that 26 of them were unexpected or surprising to the participants. Among them, 9 were developed with the baseline and 17 were derived with KGScope. This is also consistent with participants' feedback that KGScope exposed information relevant to the entities of interest, which helped participants gain an understanding of them and provided a greater opportunity to reveal unexpected insights.

Participants explored entities in groups and uncovered insights related to them: When using baseline, a total of 15 insights were reported that included groups, whereas when using KGScope, 22 insights were mentioned that included groups. For those participants who did not find commonalities of the groups with baseline, they found insights related to direct connections or commonalities between some of the entities. For example, P3 explored movies that have the keyword "bravery" and ended up

findings might occur by combining them." - P11. Additionally, 8 participants considered that the colors of the entity groups in the MRKGPanels were very helpful in distinguishing the relations and *"recalling the insights even when looking back after a while."* - P10. Separating the wrap-up workspace enabled participants to gather only informative subgraphs after a thorough exploration. By integrating the data with effective visual encodings to represent the observations, participants could summarize them efficiently.

D. Limitations and Future Work

While the results indicate that KGScope can assist users in knowledge graph exploration, it is mainly a prototype that has several limitations.

Although knowledge graph embeddings are useful for identifying peculiar triplets, the link prediction from embeddings may not be 100% accurate. However, this does not diminish the value of using peculiarity as a guidance for exploration. In fact, the results of this study suggest that embedding-based guidance based on multiple embedding models can be an effective approach. On the other hand, further research could investigate how embedding-based guidance can be integrated with other visualization techniques to enhance knowledge graph exploration and analysis.

Another limitation related to knowledge graph embeddings is that the pre-processing of the input dataset is computationally expensive. Although the interactive exploration would not be influenced if the dataset is fixed, it's worth studying ways to accelerate the process and reduce the time needed for embedding training. At this stage, we focus on demonstrating how leveraging information provided by embeddings is effective to assist human exploration of knowledge graphs.

Also, if users have particular preferences, their preferences will dominate the exploration process. Although relation recommendations are calculated based on the groups selected by users and the color mapping of peculiarities of entities in the EBViews are adjusted dynamically when filters are applied by users, they are basically data-driven. Currently, the system does not adapt to the user preferences. Acquiring user preferences through their interaction history [58] or other approaches and designing algorithms that take them into account can produce more personalized recommendations for the exploration of knowledge graphs. To comprehensively assist different user behaviors, we may improve KGScope by adapting the guidance to personal preferences.

In the user study, we evaluated the effectiveness of the guidance by comparing KGScope with a baseline system without guidance. In the future, we could adopt a more sophisticated questionnaire design, such as a recent methodology proposed by Ceneda et al. [59], to improve the measure.

X. CONCLUSION

We introduce KGScope an interactive visual exploration system with embedding-based guidance, to assist humans in exploring knowledge graphs in breadth and depth and further acquiring insights. While formulating design requirements, we take users'

needs and knowledge graphs' characteristics into account. With KGScope, users can obtain an overall understanding of the data and be guided to systematically explore the knowledge graph. At the same time, users can gather interesting observations to derive intriguing insights. Recommending relations besides entities enabled analysis from different perspectives and revealed fascinating associations. Our usage scenarios and user study demonstrate the effectiveness and generalizability of KGScope in exploring different knowledge graphs. Finally, we verify the fulfillment of design requirements and validate the effectiveness of embedding-based guidance on knowledge graph exploration in a comparative user study.

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Chao-Wen Hsuan Yuan received the master's degree in computer science from National Yang Ming Chiao Tung University. She is currently with Synology Inc., Taiwan. Her research interests include data visualization and human-computer interaction.



Tzu-Wei Yu is working toward the master's degree with National Yang Ming Chiao Tung University. His research interests include data visualization and human-computer interaction.



Jia-Yu Pan received the PhD degree from Carnegie Mellon University. He is currently with Google Inc., USA. His research interests include data mining on multimedia and graphs, clustering techniques, anomaly detection methods, and web-based services and cloud computing.



Wen-Chieh Lin received the PhD degree in robotics from Carnegie Mellon University. He is a professor with the Department of Computer Science, National Yang Ming Chiao Tung University. His research interests span several areas of computer graphics, human-computer interaction, visualization, and robotics.